

ANDREAS BIRI

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EMBEDDED WIRELESS ENGINEER

Dedicated embedded engineer specialized in wireless networking and embedded systems with a PhD in wireless sensor networks. Demonstrated expertise in designing and optimizing low-power, reliable wireless solutions using BLE, Bluetooth Classic, UWB, cellular, and proprietary protocols. Proficient in embedded C, C++20, and Python with hands-on experience in FreeRTOS, Zephyr, embOS, and bare-metal development on ARM Cortex-M. Proven record delivering impactful products, from autonomous sensor networks to medical devices and lasers, while coordinating and collaborating effectively with hardware, firmware, QA, and app teams across sites.

CAREER HIGHLIGHTS:

- PhD focused on designing sensors and protocols to deploy autonomous networks in the Swiss Alps
- Technical lead for a product improvement update of the microcontroller software in a medical device
- Team representative in critical project meetings with internal stakeholders within months of joining

EXPERIENCE

STAFF EMBEDDED SOFTWARE ENGINEER | October 2025 – present

Lightmatter | Boston, MA

Developing firmware for optical embedded systems, designing standard-compliant APIs to provide seamless laser integration and control, and collaborating on system architecture design, HW bring-up, and testing.

- Developing embedded software in C using FreeRTOS and Zephyr on ARM Cortex-M4 and M55
- Supporting board bring-up and hardware validation, writing device drivers for I2C and SPI peripherals
- Designing extensive Python APIs and HALs to boost usability for domain experts and test engineers

EMBEDDED SOFTWARE ENGINEER | August 2023 – June 2025

Sonova | Stäfa, Switzerland

Developed, specified, and coordinated embedded control applications for the wireless sub-system of an industry-leading hearing aid, focusing on wireless interactions through a proprietary wireless ASIC.

- Developed embedded software in modern C++ on a proprietary low-power microcontroller using embOS
- Designed, implemented, tested, and validated an extensive wireless communication stack using 2.4 GHz (Bluetooth Classic, BLE incl. LE Audio, and proprietary protocols for low-latency audio and data)
- Extended and optimized the interface design between embedded software and proprietary wireless ASIC
- Integrated and automated testing of lower-level wireless firmware in close cross-team collaboration
- Planned and executed months-long product stabilization phases to pass medical device certifications
- Drove participation in Bluetooth testing events to assess and improve interoperability with major vendors
- Analyzed over-the-air Bluetooth traffic and communication on SPI, I2C as well as chip-internal messaging
- Performed system testing in C# for Bluetooth Classic profiles on Android devices and LE Audio services

RESEARCH ASSISTANT (PHD) | January 2019 – July 2023

ETH Zürich | Zurich, Switzerland

Designed cutting-edge wireless protocols, developed novel sensing systems, and built and tested the designs with indoor testbeds as well as outdoor deployments in harsh alpine environments to validate their applicability.

- Developed communication schemes and wireless protocols for low-power autonomous sensor networks
- Designed and programmed custom PCBs in embedded C using FreeRTOS for analog sensing and wireless communication across various frequency bands (sub-GHz, LoRa, 2.4 GHz, UWB, cellular, GNSS)
- Built mixed-signal PCBs with multiple radios and COTS sensors connected via I2C, SPI, UART, and USB
- Published four first-author papers and presented them at first-class conferences (SenSys, IPSN, MobiCom)
- Tested designs rigorously through automated test infrastructure in indoor testbeds as well as multi-year outdoor deployments with live monitoring, remote control, and continuous data analysis

EDUCATION

Certified Professional for Software Architecture (CPSA-F), iSAQB | Zurich, Switzerland

Doctor of Sciences, ETH Zürich | Zurich, Switzerland

- Focus: Wireless sensor networks, autonomous networks, wireless protocols, low-power sensing

Master's thesis, University of California, Berkeley | Berkeley, CA

- Focus: Design of a wearable for distance measurements using UWB and Bluetooth Low Energy

Master of Science in Electrical Engineering and Information Technology, ETH Zürich | Zurich, Switzerland

- Focus: Embedded systems, advanced computer networks, wireless communication, network security

TECHNICAL SKILLS

LANGUAGES: C, C++20, Python, C#

EMBEDDED OSES / RTOS: FreeRTOS, Zephyr, embOS, bare-metal

WIRELESS PROTOCOLS: BLE incl. LE Audio, Bluetooth Classic, UWB, GNSS, cellular, LoRa, WiFi, proprietary

DEBUG TOOLS: ElliSys, RF Creations, Wireshark, Logic Analyzer, SEGGER JLink, oscilloscope, optical power meter

DEVELOPMENT TOOLS: Git, CMake, Jira, Confluence, GH Actions, Jenkins, Elasticsearch, NUnit, Catch2, VSCode

HARDWARE DESIGN: ARM Cortex-M0/M4F (Infineon PSOC6, nRF52840/54L15, STM32L/F), SemTech, uBlox, Qorvo

PROJECTS

- *LE Audio support* (Sonova): Drove the initial support of LE Audio for market-leading hearing aids. Co-designed the state machine to represent BIS sync states, implemented PACS, CSIS, and BASS services, and developed new interfaces with the wireless ASIC. Accomplished proof-of-concept within three months and successfully tested the stack at two international Bluetooth events with other major vendors.
- *Product improvement update* (Sonova): Led the technical implementation of a firmware update that improved stability and user experience for more than 1.5 million end consumers. Implemented and thoroughly tested AVRCP Volume Control and CTKD for Bluetooth Classic. Coordinated five teams across sites, systematically fixed bugs, and ensured timely firmware delivery for medical certification.
- *Autonomous networking* (ETH Zürich): Designed a protocol for low-power wireless networks that guarantees robust communication despite arbitrary node and link failure. Proved that the system yielded unique decisions with low delay and energy overhead, validated the protocol in a deployment of 23 nodes, and demonstrated its ability to re-establish efficient and reliable communication within seconds.
- *Early warning system* (ETH Zürich): Developed a novel communication scheme for remote sensor networks, combining an efficient local mesh and long-range communication via LoRa. Programmed a Python simulator to evaluate and optimize three wireless protocols and demonstrated the concept in a multi-year outdoor deployment, reducing latency by 87% and improving energy efficiency by 58x.